IILM UNIVERSITY · SCHOOL OF COMPUTER SCIENCE & ENGINEERING

Student Grade Management System

Data Structures & Algorithms — Java

Candidate: Aayush Kumar

Program: CodeZoner Virtual Internship

Duration: 4 Weeks · 29 Jul – 26 Aug 2026

Candidate & Program Overview

Candidate Details

STUDENT NAME

Aayush Kumar

ROLL NUMBER

2410030408

DEGREE

B.Tech (Computer Science & Engineering)

ACADEMIC BATCH

2024 – 2028

UNIVERSITY

IILM University, Greater Noida, U.P.

Internship Profile

PROGRAM NAME

CodeZoner Virtual Internship

DOMAIN

Data Structures & Algorithms — Java

DURATION

4 Weeks (29 Jul – 26 Aug 2026)

STUDENT ID

CZ-2026-0199

OFFER LETTER REF

CZ/OFL/CZ-2026-0199/2026

Organization Profile: CodeZoner

CodeZoner is a virtual internship platform offering structured, project-based learning for engineering students. For this internship it provided a four-week plan spanning Java development, data structures, core implementation, testing, and final submission.

Structured Curriculum

A defined four-week roadmap covering Java fundamentals, data structures, core development, and testing milestones.

Project-Based Learning

Real assignments — like the Student Grade Management System — replace passive theory with hands-on software building.

Practical Skill Building

Bridges classroom programming concepts with real development practices: version control, documentation, and delivery.

Problem Statement & Project Scope

The Challenge

Managing student records and academic grades manually becomes difficult as numbers grow. Repeated calculation of totals, percentages, and grades is time-consuming and error-prone — a simple, reliable system is needed to store records and automate these operations.

The Project Scope

- → Adding, viewing, and searching student records
- → Updating marks with automatic total, percentage & grade calculation
- → Sorting records by academic performance
- → Generating student transcripts
- → Validating input and handling invalid entries
- → Logging activity, unit/performance testing, and JAR packaging

Core Project Objectives

Core Development

OOP design via Student, StudentService & GradeCalculator classes; ArrayList-based storage with ID-based search.

Grade Automation

Automatic calculation of total marks, percentage, and final grade for every student record.

Validation & Testing

Input validation plus JUnit unit tests and a 10,000-record performance test.

Tooling & Delivery

Git/GitHub version control, JAR packaging, and a menu-driven console interface.

Technologies & Tools Used

Category	Tool / Technology	Use in Project
Language	Java 17	Application implementation
IDE	IntelliJ IDEA	Coding, debugging, and testing
Data Structure	ArrayList	In-memory student records
Version Control	Git & GitHub	Source control and documentation
Testing	JUnit 5	Unit testing of key functions

System Architecture

Main.java — Presentation Layer

Displays the console menu, accepts user input, and calls StudentService.

StudentService — Business Logic

Manages adding, searching, sorting, updating, and transcript generation via ArrayList<Student>.

Student / GradeCalculator / InputValidator — Domain Layer

Store student data, calculate totals, percentage & grade, and validate all inputs.

AppLogger — Logging Layer

Records selected activities for traceability during execution.

Result — Output

Returns calculated grades and generated transcripts back to the console.

Curriculum Timeline (Weeks 1–4)

WEEK 1

Setup & Research

Installed Java & IntelliJ, set up the GitHub repo, studied data structures, and built the Student class with basic validation.

WEEK 2

Core Implementation

Record storage, sorting, grade updates, invalid-input handling, unit tests, and a 10,000-record performance test.

WEEK 3

Advanced Features & Polish

Transcript generation, improved console UI, user testing, bug fixing, logging, and documentation.

WEEK 4

Packaging & Submission

JAR packaging, README finalization, demo recording, final testing, and project submission.

Key Features & Demonstrated Skills

Object-Oriented Java Development

Practical experience designing classes and applying OOP principles in a real application.

Data Structures in Practice

Used ArrayList for in-memory storage, plus custom search and sort logic across student records.

Automated Grade Processing

Built logic to calculate totals, percentages, and grades, then generate transcripts automatically.

Testing & Validation

Applied JUnit unit tests, a 10,000-record performance test, and input validation for robustness.

Version Control & Packaging

Used Git/GitHub for source control and packaged the finished app as an executable JAR.

Certification & Official Verification

Verification Metric	Official Record / Credential Details
Internship Title	Data Structures & Algorithms — Java Virtual Internship
Issuing Organization	CodeZoner EdTech Private Limited
Assigned Project	Student Grade Management System
Certificate ID	CERT-CZ263J7058 (Issued: 24 Aug 2026)
Offer Letter Ref	CZ/OFL/CZ-2026-0199/2026
Student ID	CZ-2026-0199
Verification	codezoner.in/verify

Certificate of Completion

Thank You!

Aayush Kumar

School of Computer Science & Engineering

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